

Algebra First-Year Practice Exam A, Part 1

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Let G be a group of order 44. Prove that
 - (a) a Sylow 11-subgroup of G is normal in G , and
 - (b) the centre of G contains an element of order 2.
2. Let $G = A \times B$ be a finite group which is the (internal) direct product of two subgroups A and B which are nonabelian simple groups.
 - (a) Show that the only proper, nontrivial normal subgroups of G are A and B . (Hint: If N is a proper normal subgroup different from A and B , consider the commutator $[N, A]$.)
 - (b) Give an example to show that (a) would be false were A and B allowed to be abelian.
3. Let $G = \text{GL}_2(\mathbb{R})$ act on $\mathbb{R}^2$ by matrix multiplication: $A \cdot \mathbf{v} = A\mathbf{v}$ for $A \in G$, $\mathbf{v} \in \mathbb{R}^2$.
 - (a) Determine the stabilizer in G of $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \in \mathbb{R}^2$.
 - (b) Determine the orbit of $\mathbf{e}_2$.
 - (c) How many orbits are there for the action of G on $\mathbb{R}^2$?
4. Let G be a group.
 - (a) Define the ascending central series
$$Z_0(G) \leq Z_1(G) \leq Z_2(G) \leq \dots$$
for G . (This is also known as the upper central series.)
 - (b) Prove that the groups $Z_i(G)$ are all characteristic subgroups of G .
 - (c) Give an example of a nontrivial group G such that the groups $Z_i(G)$ are all trivial.
5. Let $n \geq 2$ and let Z_n be a cyclic group of order n . Prove that

$$\text{Aut}(Z_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times.$$